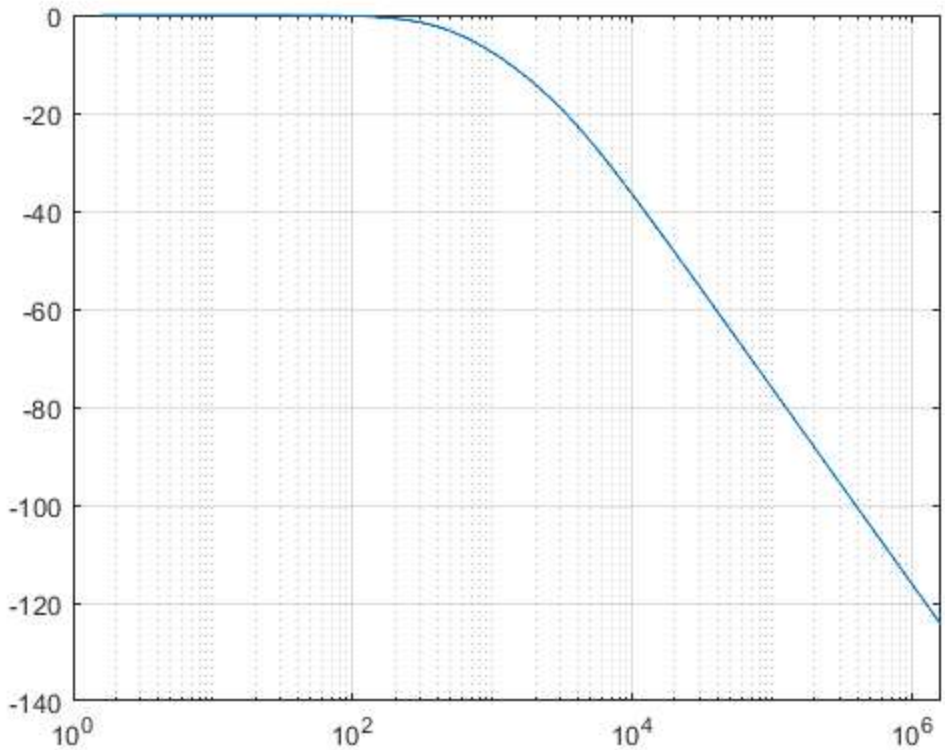
```
R1 = 200/3;
R2 = 250;
C1 = .000001; %10^-6
C2 = .000001; %10^-6
G = [1, 0, 0;
    -1/R1, 1/R1+1/R2, -1/R2;
     0, -1/R2, (1/R2)];
C = [0, 0, 0;
    0, C1, 0;
    0, 0, C2];
b =[1;
    0;
    0];

w=0:10:10e6;

semilogx((w/(2*pi)), freqresp4(G,C,b,w))
grid on
% xscale log

function F=freqresp4(G,C,b,w)

mag=zeros(3,1);
% We need to initialize vector mag, in order not
% to confuse Matlab.


for k=1:length(w)
    omega=w(k);
    % Omega is the next frequency in vector w.

    A=G+i*omega*C;
    x=A\b;
    % Vector x contains the solution of the node voltage
    % equations in complex form.

    mag=[mag abs(x)];
    % The function abs(x) computes the magnitudes of all voltages.
    % These magnitudes are stored as a new column in matrix mag.
end
% Each column of matrix mag now contains voltage magnitudes computed
% at a particular frequency. Note, however, that the first column of
% this matrix is essentially redundant. It is a vector of zeros that
% serves only as a place holder.

V3=mag(3,2:length(w)+1);
% Since we are interested only in V3, we will extract row 3 of matrix
% mag (and ignore the first column).
F=20*log10(V3);

end
```



Published with MATLAB® R2023b